

TITANIC SURVIVAL PREDICTION

MACHINE LEARNING-INTERNSHIP REPORT

Objective

- The objective of this project is to predict whether a passenger survived the Titanic disaster using machine learning techniques.
- The project utilizes passenger information such as age, gender, passenger class, fare, and family details to build a predictive classification model.

Explainability and Results

- Feature Importance analysis was used to identify the most influential features affecting survival prediction.
- Important features included Sex, Age, Fare, Passenger Class, and the engineered features.
- The trained model was saved as "titanic_model.pkl" and successfully tested using a sample inference example.

Model Development

- A Random Forest Classifier was used for training and prediction.
- The dataset was divided into training and testing sets, and the model was trained on the processed data.
- Predictions were generated on the test set and evaluated using Accuracy Score, Classification Report, and Confusion Matrix.

Data Preprocessing and Feature Engineering

- Data preprocessing was performed to improve data quality and model performance. Missing values in the Age column were filled using the median value, while missing Embarked values were replaced with the mode.
- Three new features were created:
 - Title: Extracted from passenger names.
 - FamilySize: Calculated using SibSp + Parch + 1.
 - CabinPresent: Indicates whether cabin information is available.
- Categorical variables such as Sex, Embarked, and Title were encoded into numerical values using Label Encoding.

Dataset Description

- The Titanic dataset contains passenger-related information including PassengerId, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked, and Survived.
- The target variable is Survived, where 1 indicates survival and 0 indicates non-survival.

Conclusion

This project demonstrates a complete machine learning workflow, including preprocessing, feature engineering, model training, evaluation, explainability, and model deployment. The Random Forest model successfully predicted passenger survival and provided interpretable results through feature importance analysis.